

Electricity Demand Forecasting Using Deep Learning

1. Introduction

This project develops an end-to-end **hourly electricity demand forecasting system** using historical electricity consumption data. The project starts with data preprocessing and exploratory analysis, followed by baseline forecasting methods and deep learning models such as **RNN, LSTM, GRU, and Bidirectional LSTM**. Different sequence lengths and model improvement techniques are evaluated to identify the best-performing forecasting model. Finally, the selected model is intended to be integrated into a **Flask + Bootstrap interactive forecasting dashboard**.

2. Objectives

To understand historical electricity demand patterns. To preprocess and clean hourly electricity consumption data. To perform exploratory data analysis. To establish baseline forecasting models. To develop deep learning models for electricity demand forecasting. To compare RNN, LSTM, GRU, and Bi-LSTM models. To investigate different sequence lengths. To improve model performance using different training techniques. To evaluate models using multiple forecasting metrics. To select the best-performing forecasting model. To develop an interactive forecasting dashboard using Flask and Bootstrap.

3. Dataset

The project uses the **PJM East (PJME) hourly electricity demand dataset**.

The main target variable is:

PJME_MW

The dataset contains timestamped hourly electricity consumption values. The timestamp is used to extract temporal information such as:

- Hour
- Day
- Week
- Month
- Day of Week
- Weekend

These temporal features help the forecasting models learn recurring electricity demand patterns.

4. Project Workflow

The overall workflow is:

Raw Dataset



Data Understanding



Data Cleaning



Datetime Processing



Missing Timestamp Handling



Outlier Detection & Treatment



Feature Engineering



Scaling



Exploratory Data Analysis



Baseline Forecasting



Deep Learning Models



Model Improvement



Model Evaluation



Best Model Selection



Flask Forecasting Dashboard

5. Phase 1 – Data Understanding

The first phase focuses on understanding the structure and characteristics of the dataset.

The following aspects were examined:

- Number of rows and columns
- Data types
- Timestamp information
- Target variable
- Missing values
- Duplicate records
- Basic statistical characteristics
- Electricity demand distribution

The timestamp column was converted into a proper datetime format to enable time-series analysis.

6. Phase 2 – Data Preprocessing

The preprocessing phase prepares the raw time-series for machine learning. The data is sorted chronologically and missing timestamps or missing demand values are checked before modelling.

- Chronologically sort the data.
- Check and handle missing timestamps.
- Handle missing demand values.
- Detect and treat abnormal values where appropriate.
- Create time-based features such as hour, day, week, month, and day of week.
- Create lag and rolling statistical features.

Rolling features capture recent demand behavior. A 24-hour rolling mean represents the recent daily demand level, while a 168-hour rolling mean represents the recent weekly demand level.

7. Phase 3 – Exploratory Data Analysis

Exploratory Data Analysis was performed to understand electricity demand patterns.

The analysis focused on:

Hourly patterns

Electricity demand changes throughout the day. Peak and low-demand periods were identified.

Daily patterns

Demand patterns were analyzed across different days.

Weekly patterns

Weekday and weekend consumption patterns were compared.

Monthly patterns

Seasonal variations in electricity demand were examined.

Rolling trends

Rolling averages were used to visualize short-term and weekly demand trends.

EDA helped determine that electricity consumption contains strong **temporal and seasonal patterns**, making time-series deep learning models suitable for the forecasting task.

8. Phase 4 – Baseline Forecasting

Before developing deep learning models, simple forecasting approaches were implemented as baseline models.

The four baseline models were:

1. Naive Forecast
2. Moving Average
3. Previous-Day Forecast
4. Previous-Week Forecast

The models were evaluated using:

- MAE
- MSE
- RMSE
- MAPE
- R^2
- Forecast Bias

9. Phase 5 – Deep Learning Model Development

Four deep learning architectures were developed:

- Basic RNN
- LSTM
- GRU
- Bidirectional LSTM

Different sequence lengths were investigated:

- 24
- 48
- 168

A sequence length represents the number of previous time steps provided to the model to predict the next demand value.

10. Phase 6 – Model Improvement

Phase 6 focused on improving the performance of the deep learning models.

Different techniques were investigated, including:

- Learning-rate scheduling
- Early stopping
- Different batch sizes
- RMSprop optimizer
- Dropout
- Batch normalization
- Additional layers
- Increased number of units

Phase 6 Best Model

The **GRU with sequence length 168** achieved the best overall deep-learning performance.

It achieved:

- MAE = **1134.378**
- RMSE = **1790.525**
- MAPE = **4.046%**
- $R^2 = 0.848346$

Therefore, it was selected as the final candidate forecasting model.

11. Phase 8 – Model Evaluation

The selected models are evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- R² Score
- Forecast Bias

12. Final Model Selection

Based on the Phase 6 results, the **Bidirectional LSTM with sequence length 168** was selected as the final deep-learning forecasting model.

The model achieved:

RMSE = 1805.525407

R² = 0.918346

MAPE = 4.045682%

The model captures both forward and backward dependencies within the input sequence, making it suitable for learning complex temporal relationships in electricity demand.

13. Phase 9 – Forecasting Dashboard

The final forecasting system is planned to be deployed using:

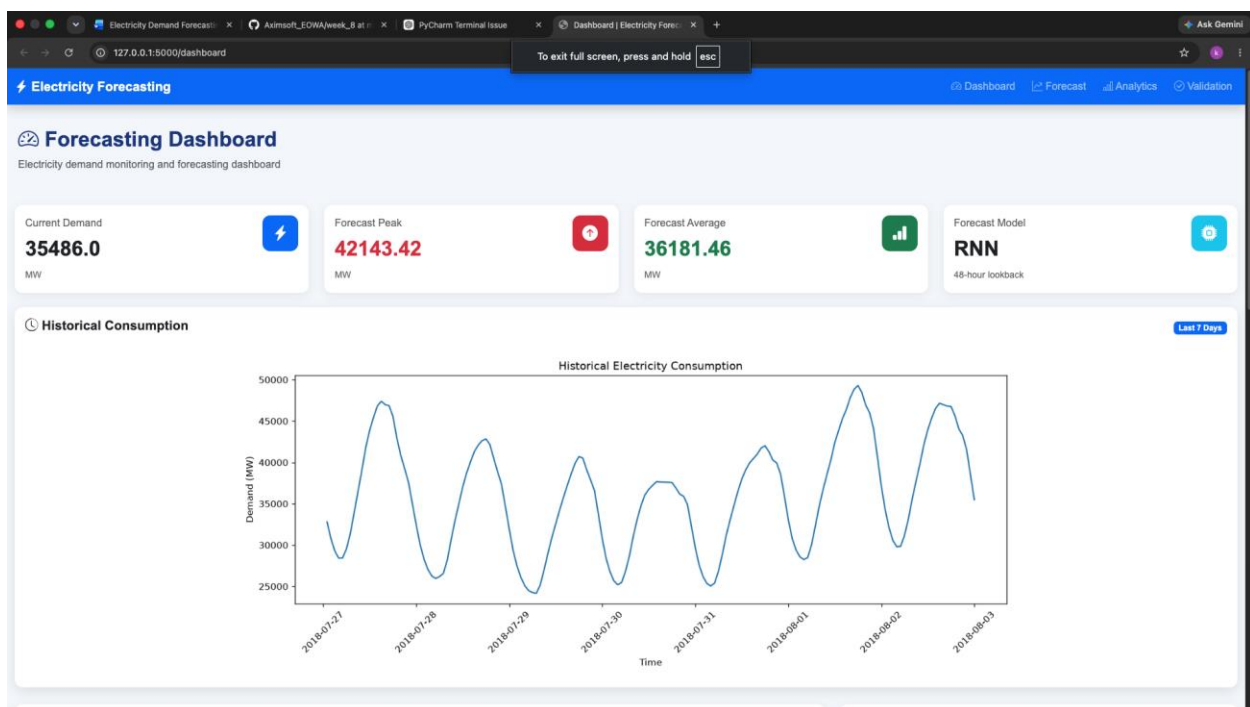
- **Python**
- **Flask**
- **Bootstrap**
- **HTML**
- **CSS**
- **JavaScript**
- **Jinja2**

The dashboard will provide an interactive interface for viewing historical electricity demand and generating forecasts.

13.1 Dashboard

The main dashboard will display:

- Current/latest electricity demand
- Historical consumption
- Forecast summary
- Selected model
- Model performance



13.2 Forecast Page

The forecast page will allow users to:

- Select the forecasting horizon
- Generate future demand predictions
- View forecast values
- Visualize future demand
- Display forecast intervals where implemented

Example:

Forecast Horizon



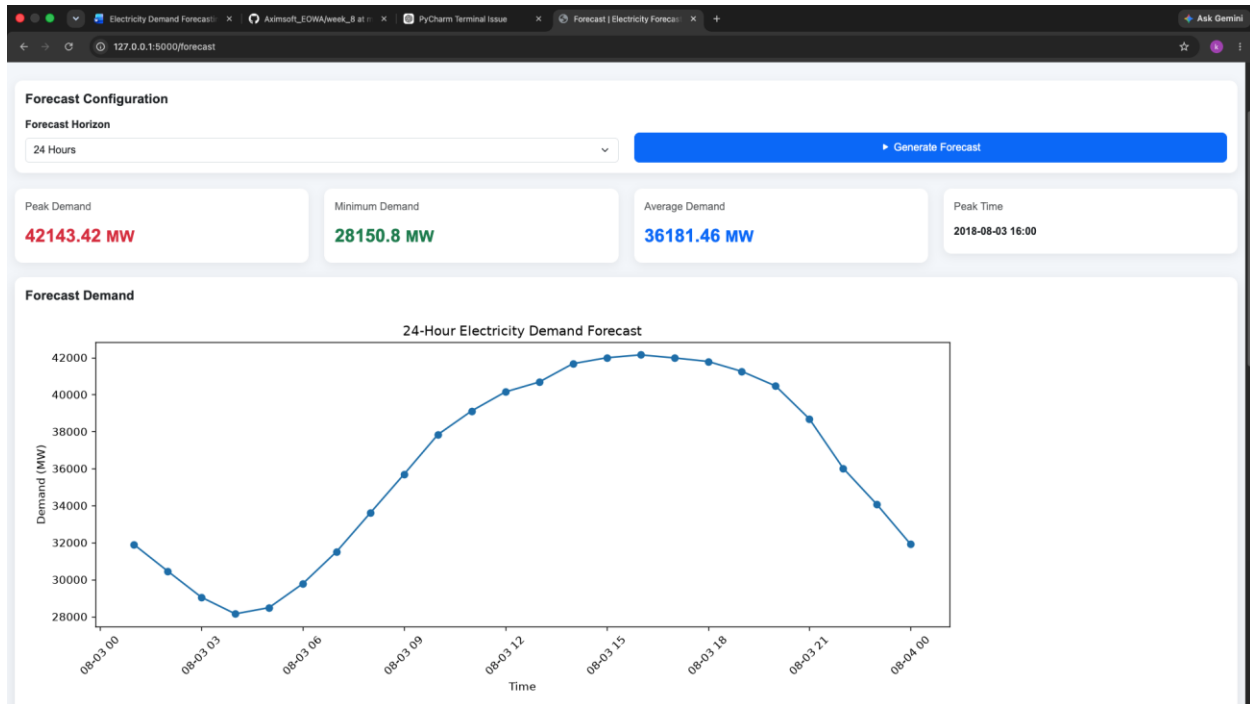
24 Hours



Generate Forecast



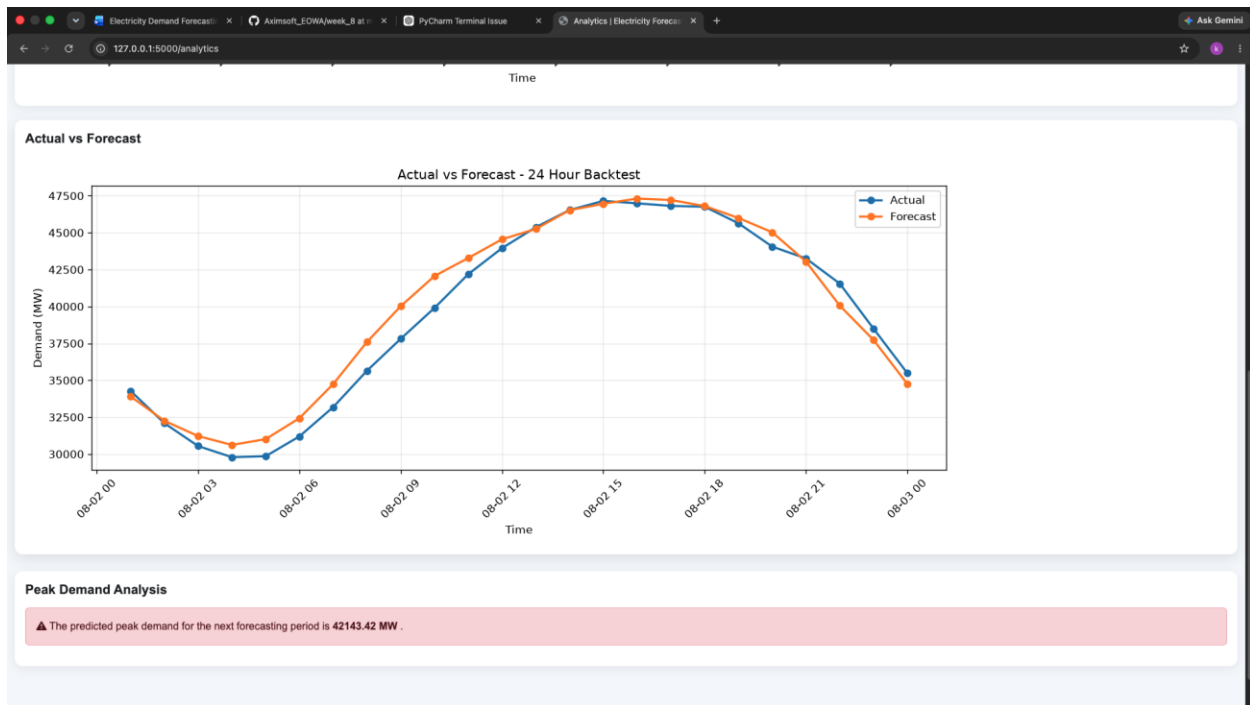
Future Electricity Demand



13.3 Analytics

The analytics page will contain:

- Actual vs Forecast
- Daily trends
- Weekly trends
- Monthly trends
- Peak demand analysis
- Forecast errors
- Model comparison



13.4 Validation

The validation page will allow the user to select:

- 30 days
- 60 days

The page will display:

- Actual vs predicted demand
- MAE
- RMSE
- MAPE
- Forecast error visualization

Electricity Demand Forecast | x | AutoSoft_ECWA/week_8 at | x | PyCharm Terminal Issue | x | Validation | Electricity Forec | x | +

127.0.0.1:5000/validation

Ask Gemini

Electricity Forecasting

DashboardForecastAnalyticsValidation

Forecast Validation

Evaluate forecasting performance on unseen validation periods.

Select Validation Period

Validation Period

30 Days

Validate Model

Validation Period

30 Days

Validation Records

720

Actual Average

29684.96 MW

Actual Peak

37600.0 MW

Model Validation Performance

Metric	RNN-48
MAE	2213.30
RMSE	3018.20
MAPE	6.80%
R²	0.7836

The validation period is kept separate from the final test period to maintain chronological forecasting evaluation.